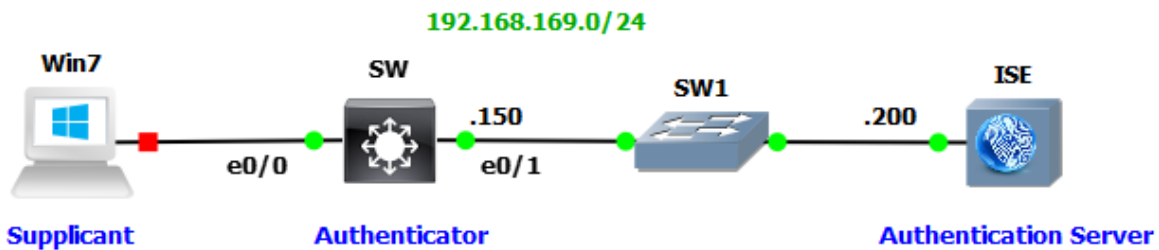


CWA Lab Topology:



Central WebAuth Configuration Procedure:

- 1- Configure the Cisco Switch for Web Authentication.
- 2- Configure the MAB to continue when user not found.
- 3- Tune the Web Authentication identity source sequence.
- 4- Configure Pre-Web Authentication Downloadable dACL.
- 5- Configure traffic redirect to WebAuth authorization profile.
- 6- Configure traffic redirect to WebAuth authorization policy.
- 7- Configure authorization policy for clients authenticated.

Configure Switch for WebAuth:

- o Enable Routing, DNS, and Default Route on Cisco Switch.
- o Enable Authentication, Authorization and Accounting.
- o Configure RADIUS Server Commands on Cisco Switch.
- o Enable HTTP/HTTPS to intercept and redirect requests.
- o Enable and Configure Change of Authorization (CoA).
- o Create & Configure Local redirection ACL in Cisco Switch.
- o Excludes traffic to ISE nodes that provide central WebAuth service.
- o Redirects all other HTTP or HTTPS traffic to Cisco ISE.
- o ACL is referenced by name in the authorization profile on Cisco ISE.
- o Apply and Configure AAA commands on host connected Port on Switch.

Enable Routing, DNS and Default Route

```
SW1(config)#ip routing
SW1(config)#ip name-server 8.8.8.8
SW1(config)#ip domain-lookup
SW1(config)#ip route 0.0.0.0 0.0.0.0 192.168.169.2
```

Configure SVI
SW1(config)#interface vlan 1 SW1(config-if)#ip address 192.168.169.150 255.255.255.0 SW1(config-if)#no shutdown
Enable AAA
SW1(config)#aaa new-model SW1(config)#aaa authentication dot1x default group radius SW1(config)#aaa authorization network default group radius SW1(config)#radius-server vsa send SW1(config)#ip device tracking SW1(config)#dot1x system-auth-control SW1(config)#radius-server attribute 6 on-for-login-auth SW1(config)#radius-server attribute 8 include-in-access-req SW1(config)#radius-server attribute 25 access-request include
Configure RADIUS Server
SW1(config)#radius server test SW1(config-radius-server)# address ipv4 192.168.169.200 auth-port 1812 acct-port 1813 SW1(config-radius-server)#key test123
Enable HTTP/HTTPS
SW1(config)#ip http server SW1(config)#ip http secure-server
Enable Change of Authorization
SW1(config)#aaa server radius dynamic-author SW1(config-locsvr-da-radius)#client 192.168.169.200 server-key test123 SW1(config-locsvr-da-radius)#exit
Create Local Redirection ACL
SW1(config)#ip access-list extended test SW1(config-ext-nacl)#permit tcp any any eq 80 SW1(config-ext-nacl)#permit tcp any any eq 443
AAA Commands on Port
SW1(config)#interface e0/0 SW1(config-if)#switchport host SW1(config-if)#authentication event fail action next-method SW1(config-if)#authentication event server alive action reinitialize SW1(config-if)#authentication host-mode multi-auth SW1(config-if)#authentication order mab dot1x SW1(config-if)#authentication priority mab dot1x SW1(config-if)#authentication port-control auto SW1(config-if)#authentication periodic SW1(config-if)#authentication timer reauthenticate server SW1(config-if)#mab SW1(config-if)#dot1x pae authenticator

Configure MAB to Continue If not Found:

- o WebAuth require a fail-open MAB authentication rule.
- o Configure it to continue with processing if the user is not found.
- o User not found authorization rule specifies redirection.

The screenshot shows the configuration for a MAB (MAC Authentication Bypass) rule. The rule is named 'MAB' and is configured to use 'Wired_MAB OR Wireless_MAB' as the identity source. The 'Allow Protocols' are set to 'Default Network Access'. The 'Internal Endpoints' tab is selected, showing the 'Identity Source' as 'Internal Endpoints'. The 'Options' section shows the following settings:

Condition	Action
If authentication failed	Reject
If user not found	Continue
If process failed	Drop

A red box highlights the 'Continue' action for the 'If user not found' condition, with a note: 'Choose Continue if user not found'. A note at the bottom states: 'Note: For authentications using PEAP, LEAP, EAP-FAST or RADIUS MSCHAP it is not possible to continue processing when authentication fails or user is not found. If continue option is selected in these cases, requests will be rejected.'

Tune the WebAuth Identity Source Sequence:

- o Configure Pre-WebAuth dACL:
- o WebAuth Redirect Authorization Profile:
- o WebAuth redirect Authorization Policy Rule:
- o Configure Authorization for WebAuth Users:

Navigate to **Policy>Policy Elements>Results >Authorization Profiles >Downloadable ACLs**
Click on **Add** to configure dACL with Name **Pre-WebAuth** allow DNS, DHCP, HTTP, HTTP and guest portal access.

The screenshot shows the configuration for a 'Downloadable ACL' named 'Pre-WebAuth'. The 'Name' field is set to 'Pre-WebAuth'. The 'Description' field is empty. The 'ACL Content' is defined as follows:

```
1 permit udp any any eq 53
2 permit udp any any eq 67
3 permit tcp any any eq 80
4 permit tcp any any eq 443
5 permit tcp any host 192.168.169.200 eq 8443
```

Navigate to **Policy>Policy Elements>Results >Authorization Profiles >Downloadable ACLs**
Click on **Add** to configure another dACL with Name **Post-WebAuth** allow any traffic.

Downloadable ACL List > Post-WebAuth

Downloadable ACL

* Name

Description

* DACL Content

1	permit ip any any
2	

Navigate to **Policy>Policy Elements>Results >Authorization Profiles**, Click on **Add** to configure Authorization Profile with Name **WEBAUTH1** Access Type: Access Accept, Choose DACL Name: Pre-WebAuth, tick Web Redirection, Choose Central Web Auth, put ACL name: test created in switch, put ISE IP statically.

Authorization Profile

* Name

Description

* Access Type

Service Template ☐

▼ Common Tasks

☒ DACL Name

▼ Common Tasks

☒ Web Redirection (CWA, DRW, MDM, NSP, CPP)

ACL Redirect

☒ Static IP/Host name

Navigate to **Policy>Policy Elements>Results >Authorization Profiles**, Click on **Add** to configure another Authorization Profile with Name **WEBAUTH2**

Authorization Profiles > **WEBAUTH2**

Authorization Profile

* Name

Description

* Access Type

Service Template ☐

▼ Common Tasks

☒ DACL Name

☐ VLAN

☐ Voice Domain Permission

☐ Web Redirection (CWA, DRW, MDM, NSP, CPP)

Navigate to **Policy > Authorization** to create **WebAuth** redirect Authorization Policy Rule.
 Create new authorization Policy named: **WebAuth** conditions: Network Access:
 AuthenticationStatus EQUALS UnknownUser and set Permissions: **WEBAUTH1**.
 Create another new policy above this named: User via WebAuth, Conditions: Network Access:
 UseCase EQUALS Guest Flow and Permissions: **WEBAUTH2**.

Status	Rule Name	Conditions (identity groups and other conditions)	Permissions
✓	User via WebAuth	if Network Access:UseCase EQUALS Guest Flow	then WEBAUTH2
✓	WebAuth	if Network Access:AuthenticationStatus EQUALS UnknownUser	then WEBAUTH1
✓	Wireless Black List Default	if Blacklist AND Wireless_Access	then Blackhole_Wireless_Access
✓	Profiled Cisco IP Phones	if Cisco-IP-Phone	then Cisco_IP_Phones
✓	Profiled Non Cisco IP Phones	if Non_Cisco_Profiled_Phones	then Non_Cisco_IP_Phones
✓	Default	if no matches, then	PermitAccess

To verify navigate to **Operations > Authentications** to see the live logs and sessions.

✓		#ACSACL#-IP-Pr	Sw1				ISE2
✓		12:34:56:78:AA: 12:34:56:78:AA:AA	Sw1	Ethernet0/0	WEBAUTH1	NotApplicable	ISE2
✓		#ACSACL#-IP-Pr	Sw1				ISE2
✓		#ACSACL#-IP-Pr	Sw1				ISE2
✓		00:0C:29:55:68: 00:0C:29:55:68:E0	Sw1	Ethernet0/0	WEBAUTH1	NotApplicable	ISE2
✓		00:0C:29:55:68: 00:0C:29:55:68:E0	VMWare-Device	Sw1	Ethernet0/0	PermitAccess	NotApplicable
✓		#ACSACL#-IP-Pr	Sw1			Profiled	ISE2
✓		00:0C:29:55:68: 00:0C:29:55:68:E0	Sw1	Ethernet0/0	WEBAUTH1	NotApplicable	ISE2

Navigate to **Administration > Identity Management > Identities > Users** click **Add** to create new user named: **support1**, Password: **Abc@12345** assign User Groups: **SponserAllAccount**

Network Access Users List > **support1**

▼ Network Access User

* Name

Status ☒ Enabled ▼

Email

► Password

► User Information

► Account Options

▼ User Groups

▼ +

From Web Authenticated system login to sponsorportal with already created username. After login create guest to users accounts to login.

Sponsor Portal - Windows Internet Explorer

https://192.168.169.200:8443/sponsorportal/

Sponsor Portal

CISCO Sponsor Portal

Username:

Password: